SOCIO-TECHNICAL POWER DYNAMICS

Technology, Power, and Whom It Serves

From Langdon Winner's "Do Artifacts Have Politics?" to power-based equality as a design goal — and what that asks of you as a technologist

Winner | ACM Code Critique | Mobile Vaani & CoRE | 17 Questions

August 2026

LECTURE ROADMAP

Four Moves, One Argument

- **Winner: Technology Is Not Neutral**

  Artifacts embed politics — in specific designs, and sometimes inherently

- **The Missing Piece in the ACM Code**

  Ethics-by-avoiding-harm isn’t enough without asking who a system is FOR

- **Mobile Vaani & CoRE Stack**

  Two real systems built to tilt power toward the poor, not just avoid harming them

- **17 Questions for Technologists**

  Realizing your own role — and organizing to change it from within

Two Views of What Technology Is

- The common-sense view: technologies are neutral tools. A hammer, a highway, a database — what matters is who uses them and why, not the thing itself. “People have politics, not things.”
- **The opposing view Winner takes seriously: some technologies embed political arrangements — patterns of power and authority — in their very design, independent of who’s using them or why.**
- Winner isn’t a technological determinist — he agrees social and economic forces shape technology. His argument is that once you’ve explained the social origins of a technology, you haven’t explained everything: the artifact itself, once built, does further political work of its own.

WINNER · ARGUMENT 1: DESIGNED-IN POLITICS

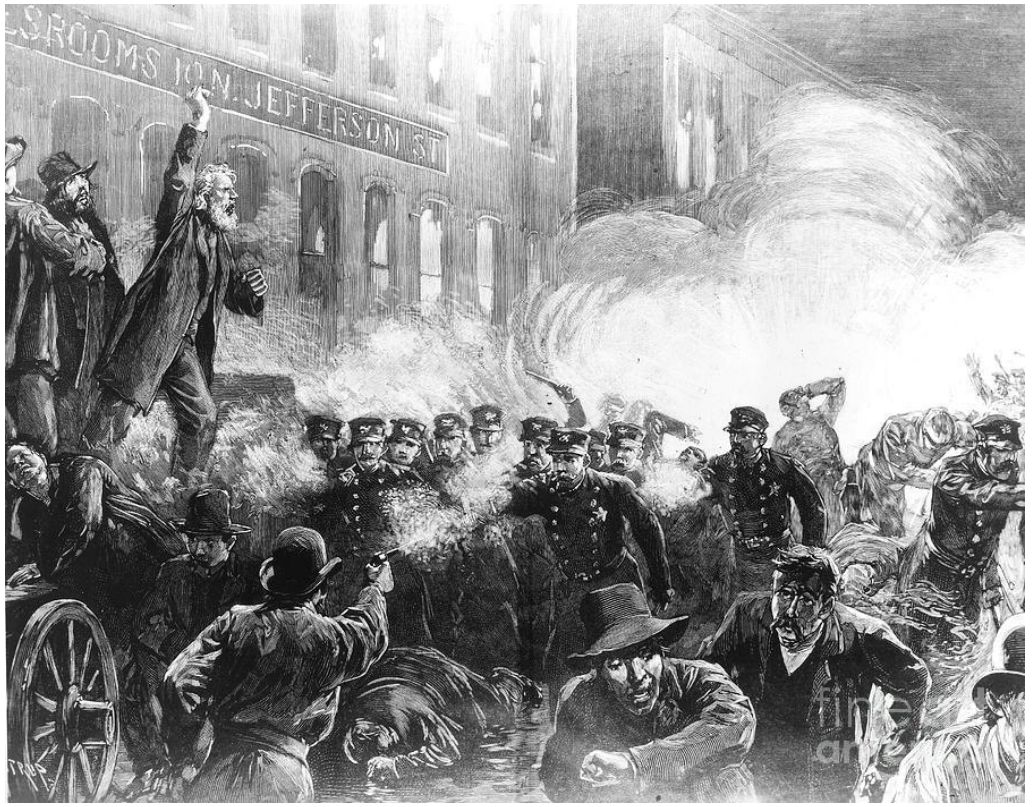
Robert Moses's Low Bridges

- New York master builder Robert Moses designed over 200 overpasses on Long Island parkways with unusually low clearance — as little as nine feet.
- **According to biographer Robert Caro, the height was deliberate: it physically prevented the 12-foot-tall public buses from using the parkways.**
- Buses were how poor and Black New Yorkers traveled. Car-owning white, middle- and upper-class residents were unaffected. The bridges quietly restricted access to Moses's public park at Jones Beach.
- The bridges look like ordinary infrastructure. They function as a durable, concrete-and-steel encoding of racial and class exclusion — outliving Moses himself.



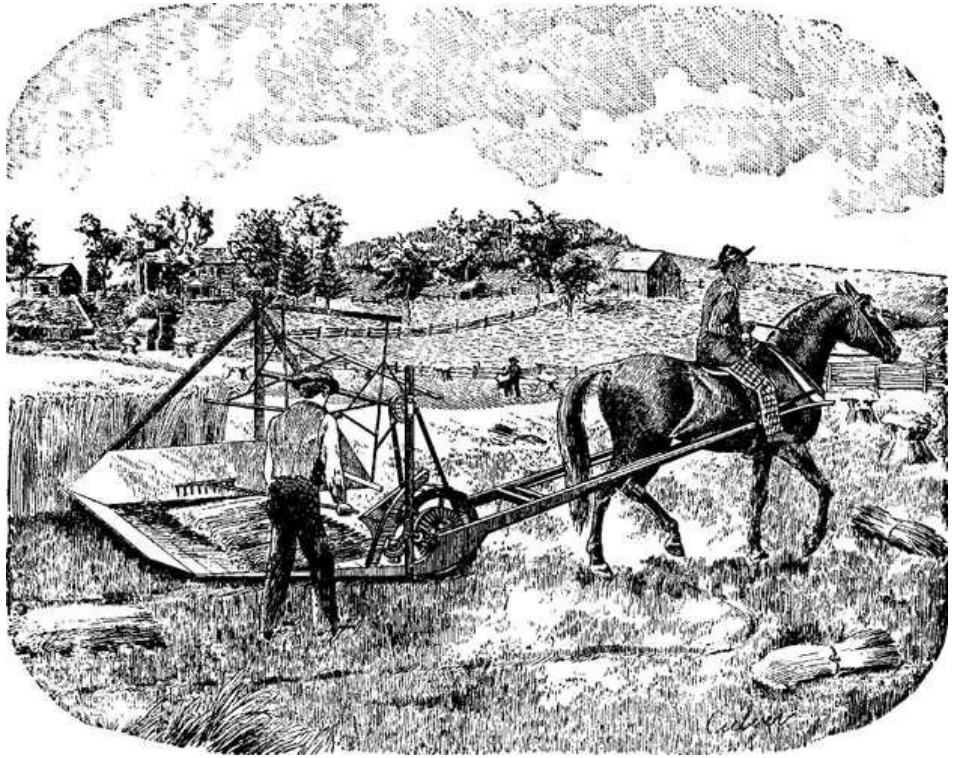
McCormick's Molding Machines

- In the 1880s, Cyrus McCormick II installed new pneumatic molding machines at his Chicago reaper factory — at the time, untested and expensive.
- **Historian Robert Ozanne's account: McCormick was in the middle of a fight with the skilled iron molders' union. The machines let him replace organized skilled workers with unskilled labor — breaking the union.**
- The machines actually produced worse castings at higher cost than the old process, and were abandoned after three years — once they had served their real purpose.
- The lesson: "efficiency" and "modernization" are not always the real reason a technology gets adopted. Sometimes the point is to reshape who has power on the shop floor.



The Mechanical Tomato Harvester

- Developed at UC-Davis from the late 1940s onward: a machine that harvests tomatoes in one pass, requiring tomatoes bred to be harder and less tasty to survive the process.
- Cost roughly $50,000 each — affordable only to large operations. Between the early 1960s and 1973, the number of California tomato growers fell from ~4,000 to ~600; an estimated 32,000 farmworker jobs were eliminated.
- **Crucially: no one has shown this was a plot. Public university researchers, agribusiness funding, and land-grant college incentives simply aligned, decade after decade, toward large-agribusiness interests — without any single actor intending to concentrate the industry.**
- This is Winner's key refinement: political effects don't require a conspiracy. A technology's development can be "thoroughly biased in a particular direction" through the ordinary, undramatic accumulation of who funds, designs, and adopts it.



When the Technology Itself Demands a Politics

- A stronger, more troubling claim: some technologies are compatible with only one kind of political arrangement, regardless of who builds or owns them.
- **Winner's clearest case: the atom bomb. Its lethal, catastrophic properties demand centralized, rigidly hierarchical, closed control — there is no democratic or decentralized way to manage a nuclear arsenal safely.**
- He revisits an 1872 debate: Engels argued that factories, railways, and ships all require strict hierarchical authority as a technical necessity, independent of capitalism or socialism — against anarchists who thought authority itself could simply be abolished.
- Winner's own position is more careful: authoritarian technologies are real and often what looks like a technical requirement is really an unexamined political choice wearing the costume of necessity.

Technology Is Not Neutral

- **Winner's summary claim: "The things we call 'technologies' are ways of building order in our world" — they settle, often for generations, how people will work, travel, communicate, and relate to one another.**
- The moment of greatest flexibility is the first time a technology is introduced — before choices harden into infrastructure, investment, and habit. After that, alternatives quietly disappear. The Collingridge dilemma!
- Designers cannot be left off the hook entirely!
- *Why this matters: Frameworks we are discussing — the ACM Code, GDPR, HIPAA — evaluates technology by its stated use and its harms. Winner asks a prior question: what relationships of power does the design itself lock in, before anyone even uses it?*

Two Questions Ethics Codes Conflate

- **Q1 — Who is your product or service meant to benefit? Answering this requires clarity about the system's actual objectives.**
- **Q2 — Is somebody being harmed by your product or service? This can be answered without ever clarifying what the system's goals are — you can patch a harm without ever asking who the system was built for.**
- The claim: the ACM Code of Ethics and Professional Conduct is built almost entirely around Q2 — avoid harm, be honest, be fair, respect privacy — and says very little about Q1.
- This connects directly to Winner: if a technology's design already encodes a politics (Section A), then a code of ethics that only screens for harm — after the fact, on the surface — will miss exactly the kind of designed-in power shift Winner describes.

A Ship Without a Compass

- Distinguish between: **Instrumental values** are preferred modes of behavior — honesty, fairness, responsibility. **Terminal values** are the actual end-states being aimed at — equality, welfare, a world at peace.
- Amartya Sen makes such a distinction for development: instrumental freedoms (e.g. market participation, political expression) vs. constitutive freedoms that need no further justification (e.g. freedom from starvation, freedom for political participation) — and famously used this to critique Rawls's "veil of ignorance" for saying too little about what outcomes actually result.
- **"Do no harm" guardrails are instrumental values only. Seth's metaphor: instrumental values alone are like a ship without a compass — they can take you to many different destinations, not all of them desirable. Only a terminal value tells you which direction to steer.**

Three Systems With Unstated Real Goals

Facebook News Feed

Publicly framed around user benefit; externally studied to actually optimize for engagement — which amplifies sensational content and consolidates echo chambers. The ethical problem isn't that the goal is unclear — it's that ad-revenue maximization was chosen over pluralistic dialogue, and never named as the trade-off it is.

Aadhaar (India's biometric ID)

Stated goal: reduce welfare leakage via biometric authentication. Researchers (Khera and others) find leakage is mostly from documentation/enrollment barriers, not identity fraud — so biometric failures instead exclude the rural poor and elderly from benefits they're entitled to.

VC-funded agri-tech startups

Framed as improving farmer livelihoods through precision agriculture. Critics argue many function as data-extraction agents of surveillance capitalism — nudging toward mono-cropping and land consolidation that increases farmer precarity rather than reducing it.

What the ACM Code Says — and Doesn't

- Section 1 (General Ethical Principles): every point except 1.1 addresses means — avoid harm (1.2), be honest (1.3), be fair (1.4), respect privacy (1.6) and confidentiality (1.7). Even 1.1 ("contribute to society and human well-being") states a goal without saying which specific outcomes count.
- Section 2 (Professional Responsibilities): also entirely means-focused — quality of work, competence, review, risk evaluation, public awareness.
- Section 3 (Leadership): builds on the same means, plus 3.7's attention to systems that become societal infrastructure — still without naming which end-states leadership should be steering toward.
- **Seth's point is not that the Code is wrong — it's that means without ends leaves "benefit of society" permanently undefined, which makes it easy for any project, including disempowering ones, to claim compliance.**
- **The proposed fix: computing needs an explicit terminal value. Seth's argument (extending Tim Unwin, Iris Marion Young, Amartya Sen) is that this terminal value should be power-based equality — removing the unjust structures that reproduce inequality, not just avoiding harm to individuals.**

Power, Defined: Types and Levels

Four Types of Power

- Power-over — domination; the negative type.
- Power-to — individual agency and capability.
- Power-with — collective power through solidarity.
- Power-within — self-esteem/confidence, a precondition for exercising the other three.

Three Levels Where Power Operates

- Resources — wealth, knowledge, position: who has direct power-over whom.
- Rules — formal and informal structures: law, bureaucracy, caste, patriarchy.
- Ideology — how the rules get legitimized: media framing, cultural common sense.
- Technology, per Winner, can encode itself at any of these three levels.

CASE: MOBILE VAANI

Voice, Not Text or Apps, as the Access Point

- Mobile Vaani (MV) is a voice-based (IVR) participatory media platform built for rural and low-literacy communities across India — no smartphone, app literacy, or even reading required to participate.
- Design choice as power choice: most digital platforms require literacy, data plans, and smartphones — which systematically excludes exactly the population MV is built for. Voice access is itself the first power-shift: it changes who can even be a speaker, not just a listener.
- **Content and moderation are managed by a hybrid online-offline model: an offline network of community volunteers governs what the online platform carries, ensuring the technology stays embedded in — and accountable to — the communities using it, rather than being run top-down by engineers who've never met a user.**
- When field feedback revealed a caste/gender skew in who was volunteering, management actively diversified the volunteer base — an example of ongoing socio-technical management, not a one-time design fix (echoing the "limits of design" argument).

CASE: MOBILE VAANI

Shifting Power in Practice

Agonistic intermediary with the state

During COVID-19 lockdowns, MV escalated citizen grievances about welfare/ration access to government departments while also carrying official announcements — not opposing the state, but forcing it to be responsive: power-to and power-with in action.

Anonymous corruption reporting

Editorial policy evolved specifically to let users report corruption without fear of retaliation — a direct response to real-world risk discovered only after deployment, not anticipated at design time.

Financial-fraud and welfare-delivery studies

Campaigns examining India’s direct benefit transfer pipeline and digital financial services surfaced fraud and knowledge gaps rural users faced — then fed those findings back to government departments and providers, turning MV into a genuine public sphere between citizens and institutions.

CASE: CORE STACK

What Communities Actually Do With the Data

- CoRE Stack is a digital public infrastructure for climate adaptation and natural resource management, built by IIT Delhi's ACT4D group with a wide network of civil-society and implementation partners.
- **Understanding and collectivizing: geospatial data (land use, water bodies, groundwater stress, cropping patterns) is presented back to communities at the village level so they can see their own landscape's condition clearly — turning individual, fragmented awareness into a shared basis for collectively organizing around its regeneration and sustainability.**
- **Accountability through a digital trail: community demands generated this way are documented and submitted as Detailed Project Reports (DPRs) to Gram Sabhas and government schemes like MGNREGA — creating a traceable record communities can point back to when following up with officials, rather than a verbal request that leaves no institutional footprint.**
- More scientific planning: combining satellite/hydrology data with ground-truthed community knowledge lets even small, under-resourced organizations produce technically credible NRM plans — without needing to hire outside hydrology experts — while keeping the underlying evidence contestable and open to correction by the community itself.

Evidence the Power Shift Is Real, Not Rhetorical

- Report-preparation time cut by roughly 70%; 40% of stewards report actively factoring equity into planning; 60% of partner organizations observed increased women's participation.
- **Caste-disaggregated data on water, land, and livestock access has repeatedly surfaced disparities that were previously invisible or undiscussable — e.g. in Odisha, mapping revealed migrant households lacked water access while landed households had adequate infrastructure, redirecting planning priorities.**
- A related MGNREGA study (Mittal, Sheth & Seth, ICTD 2024) used government MIS data plus field visits in Bihar and Jharkhand to pinpoint caste-based inequity in India's employment guarantee scheme down to the panchayat level — turning a diagnosis into an actionable data product for accountability.
- Steward testimonies describe a shift in identity: from passive data collectors to trusted institutional bridges between community and government — power-within and power-with, not just power-to.

Why Put This Responsibility on Technologists?

- The obvious alternatives — government regulation, informed consumer choice — both have real limits with digital technology specifically.
- Regulation lags: lawmakers often can't keep pace with what they're regulating, and sometimes state and corporate surveillance incentives simply align (e.g. post-9/11 security tech) rather than conflict. Even GDPR, a strong law, has provisions that are difficult to audit or enforce in practice.
- Consumer choice is blunted too: preferences are shaped by advertising and by capitalism's individualist framing (which foregrounds cost/efficiency over fairness/rights), and paid technologies reflect the preferences of paying consumers, not citizens broadly.
- **Conclusion: technologists cannot fully outsource responsibility to law or to the market. They are, structurally, the people closest to where the actual design and deployment choices get made.**

Organizations Are Embedded in a Self-Serving System

- **The division of labor creates a "moral buffer": engineers optimize for correctness, designers inject values only if the org prioritizes them, and this distributed structure lets any individual disclaim responsibility for downstream harms.**
- Private firms face investor-driven pressure to scale fast, with financial metrics ("users-per-employee") substituting for real impact metrics; states operate under a "high-modernist" faith that technology solves social problems, often obscuring shortcomings (Aadhaar/India Stack, again, as the case study).
- Even dedicated ethics functions (Chief Privacy Officers, ethics boards) often become compliance theater — present on paper, disconnected from what actually ships.
- Regulatory gaps compound this: legal responses to new technology (IP, privacy, speech) chronically lag because lawmakers often don't understand how technical architecture itself embodies values — "code is law," in Lessig's phrase, whether or not anyone wrote the law to say so.
- **The upshot: none of this is fixed by trying harder as an individual designer. It requires understanding — and organizing against — the structural incentives themselves.**

A Workers' Inquiry, Not a Checklist

- Seth's 17 questions are explicitly framed as a "workers' inquiry" — a method for building collective self-understanding among workers about their own conditions of labor, not a compliance form to fill out alone.
- Marx, 100 questions for workers (1880): https://www.marxists.org/archive/marx/works/1880/04/20.htm
- They move in one continuous arc: from noticing your own work's effects → finding your voice inside your organization → understanding why that voice might be blocked → deciding whether and how to act beyond your organization.
- *The next four slides group all 17 questions along that arc.*

See Your Own Work Clearly

1. How does your work ultimately enable the applications, services, and products that people actually use as end-users?

2. Is there a way for you to interact with those end-users — or otherwise understand how they're affected by what your work makes possible?

3. Are the outcomes positive for some people and negative for others?

4. If you don't like some of those negative outcomes — does that bother you?

Find Your Voice Inside the Organization

5. Can you change anything in your own work — or get your team, or your organization, to adopt new practices, or push changes upstream/downstream in the supply chain?

6. Are there others in your organization who share your viewpoint?

7. Do you and like-minded colleagues have real pathways to put that viewpoint across?

8. Would it actually be heard and acted on by management?

9. If it would be dismissed or ignored — do you understand the broader political economy well enough to explain why?

Understand the Political Economy

10 Could your organization's management have made the change but chose not to — or were their hands tied by wider economic/competitive pressures?

11 If change was feasible but not taken, can you think of ways to gain more power in organizational governance — IP/licensing terms, or collectivizing with colleagues?

12 If change wasn't feasible because of imperatives from the wider systems of state and markets — would you want those systems to run differently?

13 Do you think you can affect that kind of change within a democracy? Are you aware of social movements, in your country or elsewhere, pushing for it already?

Act Beyond the Organization

14 Would you lend support to those movements — by proposing new regulation, building alternate systems, or monitoring existing ones for accountability?

15 Is there recognition in wider society of the need for this kind of change, so that laws or regulations have a real chance of emerging?

16 If not — can you help build that recognition, through user engagement or campaigns where your technical knowledge is a genuine asset?

17 Or do you feel this isn't your job? If so — what do you think should be done instead to ensure technology produces desirable outcomes?

Marxist Humanism as the Grounding Argument

- For Marx, social relationships arise from relations of production and consumption. Positive relationships create genuine use-value for consumers through non-coercive production; deviations create negative relationships and alienation — workers losing control over their labor and its outcomes.
- **Seth's extension: technologists are workers too, producing technology through their own labor — and when that labor is channeled toward surveillance, engagement-maximization, or control rather than genuine use-value, technologists experience the same alienation Marx described for factory labor.**
- This reframes the 17 questions: they aren't only an altruistic checklist for helping others. Acting on them is presented as being in technologists' own self-interest — reclaiming their own humanism, not just performing duty to a distant "end user."

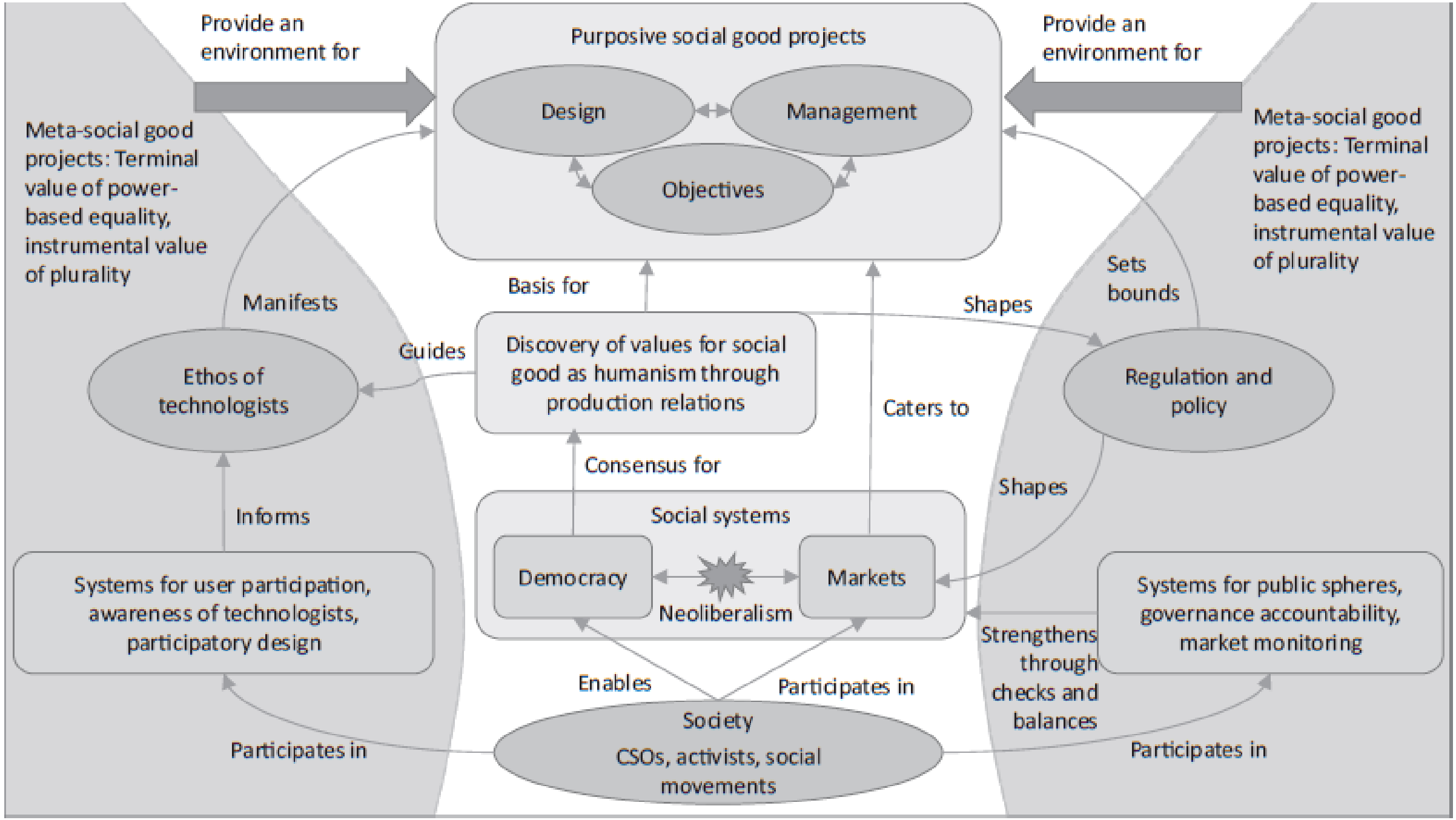
Architect or Bee? Mike Cooley and the Lucas Plan

- Cooley's title borrows a distinction Marx draws in Capital between human and animal labor: even the worst architect differs from the best bee in one crucial way — the human imagines the structure before building it, while the bee only ever executes an inherited pattern.
- **Cooley's warning: as computers and automated systems absorb more of workers' judgment and decision-making, skilled workers risk being turned from architects into bees — executing what a system dictates rather than exercising foresight over their own labor.**
- The Lucas Plan (1976) is proof that workers can still choose to act as architects. Facing mass layoffs at Lucas Aerospace (a UK arms manufacturer), shop-floor workers didn't just protest — they used their own engineering expertise to draft a 150-product "Corporate Plan" for socially useful alternatives (kidney dialysis machines, hybrid engines, wind turbines), built from shop-floor surveys run by their own combine committee.
- **Management and government ultimately rejected the plan — but that isn't the point Cooley draws from it. The Lucas workers proved that technologists are not just "objects of history" executing decisions made above them — they can be subjects capable of imagining and proposing their own future, exactly as architects do.**

"Work is a daily search for meaning as well as bread; for recognition as well as cash; for astonishment rather than torpor; in short for a sort of life, rather than a Monday to Friday sort of dying" — Frances O'Grady, General Secretary, Trades Union Congress

A Better World: What Success Looks Like

- Already visible precedents: co-determination movements demanding worker seats on corporate boards, unionization efforts (e.g. the Alphabet Workers Union), platform cooperatives, ethical-source software licenses, and appropriate-technology design that ordinary users can actually understand and control.
- **The destination Seth describes: technologies understood and controlled by their users — governed by a digital commons or democratic coalition of local institutions, not by distant corporations — with technologists embedded as part of the community, not external vendors to it.**
- The underlying values shift: cooperation over competition, power-based equality over control and domination, pluralism over closed echo chambers — with technologists, citizens, and consumers all pulling toward the same terminal value.


Provide an environment for
Purposive social good projects
Design
Management
Objectives
Provide an environment for
Meta-social good projects: Terminal value of power-based equality, instrumental value of plurality
Meta-social good projects: Terminal value of power-based equality, instrumental value of plurality
Manifests
Basis for
Shapes
Sets bounds
Guides
Ethos of technologists
Discovery of values for social good as humanism through production relations
Regulation and policy
Caters to
Consensus for
Shapes
Informs
Social systems
Democracy
Neoliberalism
Markets
Systems for user participation, awareness of technologists, participatory design
Systems for public spheres, governance accountability, market monitoring
Strengthens through checks and balances
Enables
Participates in
Participates in
Society
CSOs, activists, social movements
Participates in

SUMMARY

From Winner to Praxis

Argument	What it establishes
Winner — artifacts have politics	Design choices, even “neutral” ones, encode power relationships — sometimes deliberately, sometimes structurally
Seth — the missing clarity in the ACM Code	Avoiding harm (means) is not the same as defining who a system is for (ends) — codes need a terminal value
Power-based equality as that terminal value	Evaluate any technology by whether it shifts power-over/to/with/within toward or away from the weak
Mobile Vaani & CoRE Stack	Concrete, working proof that design and management choices can tilt power toward the poor, not just avoid harming them
17 Questions	A method for technologists to locate their own agency inside organizations shaped by capitalism and neoliberalism, and act on it

FURTHER READING

Student Reading List

Langdon Winner, "Do Artifacts Have Politics?"	*Daedalus, 1980.*	The full argument, with more examples than we could cover today.
Aaditeshwar Seth, "The Missing Clarity Between Ends and Means in the ACM Code of Ethics"	*ACM Interactions, 2023.*	The terminal-vs-instrumental values critique in full.
Aaditeshwar Seth, "Seventeen Questions for Technologists to Build a Better World"	*2022.*	All 17 questions, with full argumentation and citations.
Aaditeshwar Seth, Technology and (Dis)Empowerment: A Call to Technologists	*Emerald Publishing, 2022.*	The full book — Ch. 4 (Mobile Vaani, Aadhaar), Ch. 6 (power theory), Ch. 8 (17 questions), Ch. 11 (Agristack).
Aaditeshwar Seth, "Mobile Vaani: A Voice-based Community Media Network for Social Development"	*2022.*	A focused case-study article on Mobile Vaani alone.
CoRE Stack / ACT4D project documentation	*act4d.org, 2025–2026.*	Live project blogs on Commons Connect and the micro-watershed registry.